

Question.1

Ans $\Rightarrow$ Deep learning is a type of machine learning that uses neural networks with many layers. It can learn patterns from data automatically without needing humans to manually choose every feature. Traditional machine learning usually needs feature extraction before training, but deep learning can ~~take~~ work directly on raw data like images, text or speech.

Question.2

Activation function are used in neural networks to add non-linearity. This helps the model learn complex relationships instead of only simple straight-line patterns. Examples of activation functions are ReLU, sigmoid etc.

Question.3

Backpropagation is the training method used to improve a neural network. It calculates the error using the loss

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functions and then finds gradients for the weights. These gradients show how the weights should be changed. Using gradient descent, the model updates its weights again & again and again to reduce error.

Question - 4

Overfitting happens when a deep learning model learns the training data too closely and does not perform well on unseen data. It memorizes examples instead ~~on~~ of learning general patterns. It can be ~~related~~ reduced using dropout, regularisation, early stopping, ~~data augmenta~~ data augmentation, reducing model size.